

EOL_Instructions_for_CR5CP

CR5CP 下线检测指导规范



END OF LINE



BOSCH

部门 Department	CC-DA/EDA-CN	系统 System	CR5CP EOL
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1 更改记录 Modifications

编号 No.	版本 Version	日期 Date	修订人 Reviser	章节 Chapter	修改描述 Description of modification
1	1.0	2019/07/05	DU Yu/ CUI Yan/ ZOU Yi/ DING Yaqiang	all	初版 Initial version
2	1.1	2019/11/11	ZOU Yi/DU Yu	5	Modified process and plant requirement
3	1.2	2020/03/24	DING Yaqiang	6	更新流程图及操作步骤 Flowchart and operation process updated

2 参考文档 Reference Documents

No.	Version	Document
1	1.0	0265K60XXX_CR5CP_TCD_Part1_HW_Functions
2	1.0	0265K60XXX_CR5CP_TCD_Part2_HW_Characteristics.pdf
3	1.0	0265K60XXX_CR5CP_TCD_Part3_HW_Handling.doc
4	1.0	0265K60XXX_CR5CP_TCD_Part4_HW_Testing.doc
5	1.0	二维/三维图纸 2D / 3D drawings
6	1.0	客户诊断规范 Customer diagnostic specification
7	1.0	生产启动支持检查表 Launch management checklist
8	1.0	客户车辆配置参数 Customer Key information

3 缩写表 Abbreviations

OEM	原始设备制造商 O riginal E quipment M anufacturer
QM	质量管理 Q uality M anagement
EOL	下线检测 E nd O f L ine
DTC	诊断故障码 D iagnostic T rouble C ode
ECU	电子控制单元 E lectronic C ontrol U nit
CR5CP	第五代角雷达增强版 C orner R adar G eneration 5 C ar P lus

4 概述 General

本文档用于对生产线上雷达的安装进行指导。

以下前提条件适用于整个文档

- 毫米波雷达CR5CP在送往主机厂产线前已进行了预校准；
- 主机厂负责支架设计和安装精度；
- 毫米波雷达CR5CP安装在符合规范的覆盖件后；
- 主机厂将选用静态主动式雷达校准方式；
- 主机厂需保证在覆盖件的安装不会与雷达相干涉。

This document is to be used as a guideline for the calibration of CR5CP on product line.

The following assumptions are made throughout this document:

- CR5CP is delivered preassembled and pre-aligned to the OEM plant;
- The OEM is responsible for bracket design and installation accuracy;
- The CR5CP is mounted behind a validated cover;
- The OEM should follow static active alignment procedures;
- The OEM should ensure the cover installation will not conflict with CR5CP.

5 下线检测操作步骤 EOL process step description

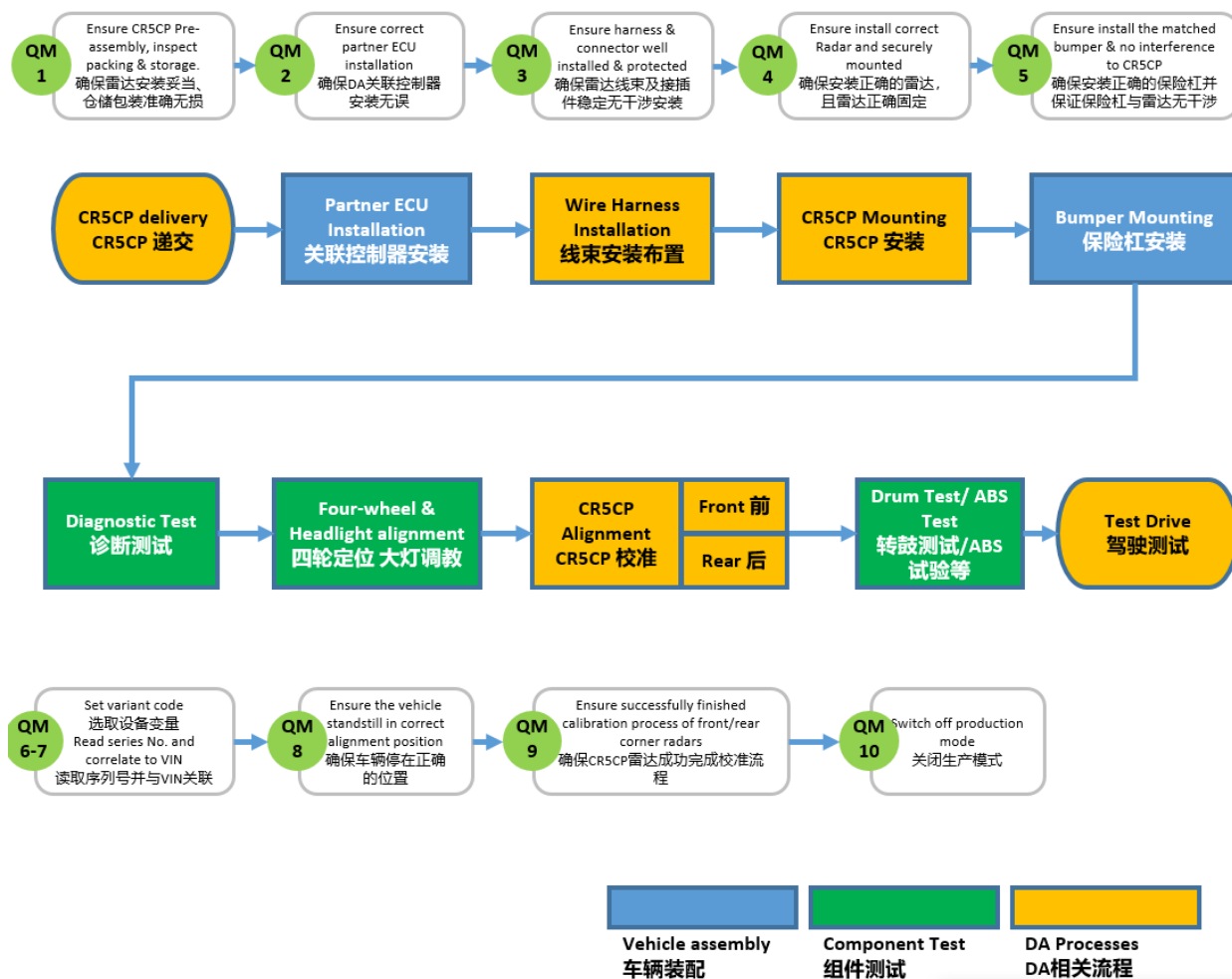


图 1: 下线检测操作步骤概览

Figure 1: EOL process overview

CR5CP 在生产线的安装及测试的推荐步骤如流程图所示，这些步骤将依据主机厂实际生产线的工位顺序进行修改。雷达校准工位可复用四轮定位工位，以保证行驶轴线符合雷达校准要求。

如复用四轮定位工位会导致重大产线变更，则必须保证车辆到达雷达校准工位前已完成四轮定位。此时在雷达校准工位上，可采用对中仪以车身物理轴线为基准进行对中，但需保证对中仪公差与四轮定位仪公差的累积值符合后文提及的校准要求的公差范围。

The main EOL steps recommend for CR5CP calibration and test are shown in the flow chart above. These steps could be modified according to the actual production sequence of the OEMs line. Succeed four wheel alignment is a precondition for CR5CP calibration, we propose to set the CR5CP calibration station in the same station of four wheel aligner to make sure driving axis meets calibration requirement.

If an individual CR5CP calibration station is considered, then the centering machine could use vehicle physical axis as reference, but the tolerance of centering machine in combine with the tolerance of four wheel aligner must follow the alignment specification mentioned in following sub-sections.

5.1 校准的目的 Purpose of alignment

校准的目的是通过确定雷达天线轴线与车辆行驶轴线在水平及垂直方向上的偏差，再通过算法补偿使二者重合或在一个允许的范围，保证配装雷达车辆的下线后相关功能能够正常使用。

The alignment has the purpose to determine the horizontal and vertical deviation angle of the radar antenna axis to the alignment reflector and thus to the driving axis of the vehicle, and then compensate the deviation angle by a certain algorithm.

5.2 校准工位 Alignment station

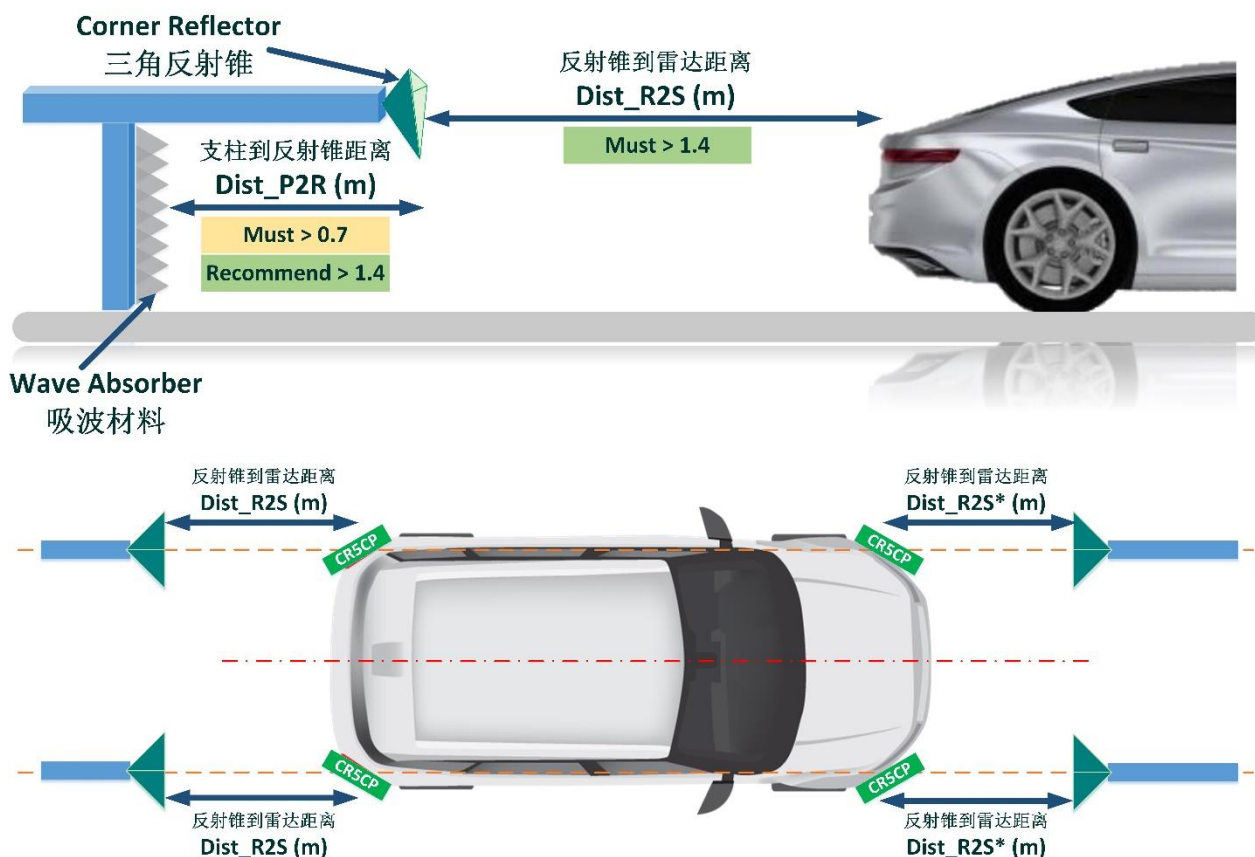


图 2：校准工位概览

Figure 2: Alignment station overview

一般的雷达校准平台如图 2 所示，包括被校准车辆、校准设备（反射锥）以及校准控制电脑等。对设备、环境的具体要求，将在后文详细描述。

Basic alignment station setup can take Figure 2: Alignment station overview as reference, generally include vehicle, alignment equipment (corner reflector) and alignment computer. The detailed requirements will be illustrated in following chapters.

5.3 校准的要求 Requirements for radar alignment

5.3.1 校准位置建议（角反射器的位置）

Recommended requirements for the alignment location (corner reflector)

雷达到角反射器的距离 Distance between sensor and corner reflector	1.41 m – 1.8 m	
角反射器到角反射器立柱的距离 Corner reflector to standing column	> 0.7 m (at least至少) > 1.4m (recommended推荐)	
角反射器在Y方向上与后轴两侧车轮接地点中心的距离 y-distance from corner reflector to rear axle center	左前 Front Left	xx mm (同雷达安装位置) (similar to radar mounting position in y)
	右前 Front Right	
	左后 Rear Left	
	右后 Rear Right	
角反射器在X方向上与车辆行驶轴线的夹角 Angle between corner reflector normal and driving axis	左前 Front Left	0° - 0°
	右前 Front Right	0° - 0°
	左后 Rear Left	0° - 0°
	右后 Rear Right	0° - 0°

5.3.2 对角反射器的要求 Requirements for corner cube corner reflector

高度height	近似雷达高度 Similar to radar height Min: xx mm Max: xx mm
材料 Material	金属Metal（推荐材料：铝） (aluminium recommended)
大小Size	10 dBsqm
位置 position	+/- 10mm
反射器边缘及底座 Plate edges and corner reflector stand	反射器边缘及底座不应反射雷达波（包括多路反射）

The mirror edges and corner reflector stand shall not reflect back to the sensor (including multipath)

注意：雷达中心须指向反射器的正中心

Note: the sensor must position in the center of the corner reflector.

反射目标根据雷达反射截面积应在如下范围 $RCS = (10 \pm 3) \text{ m}^2$. 对于如下频率(60 GHz, 77 GHz, 90 GHz), 10 m^2 可以代表 95% 的机动车。

The reflection target is defined by a RADAR Cross Section $RCS = (10 \pm 3) \text{ m}^2$. For today known frequencies (60 GHz, 77 GHz, 90 GHz), 10 m^2 represents at least 95 % of all vehicles driving on motorways.

反射目标如下图所示：

Aspect of test target should be as shown in below Figure.

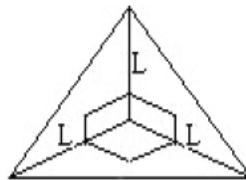


图 3：角反射器示意图

Figure 3: Corner reflector

角反射器反射截面积公式：

RCS formula of corner reflector:

$$RCS = (4 \times \pi \times L^4) / (3 \times \lambda^2)$$

这里 λ 是波长， L 是反射器的内边边长。

where λ = wavelength, L is the length of a side of a radar test corner reflector.

对于CR5CP雷达， L 推荐采用78mm.(误差为正负0.5mm)

For CR5CP, where L is recommend 78mm in principle (tolerance is $\pm 0.5\text{mm}$).

The corner in the cube should be as sharp as possible.

角反射器中心应尽可能尖锐。

5.3.3 环境条件建议 *Recommended requirements for ambient conditions*

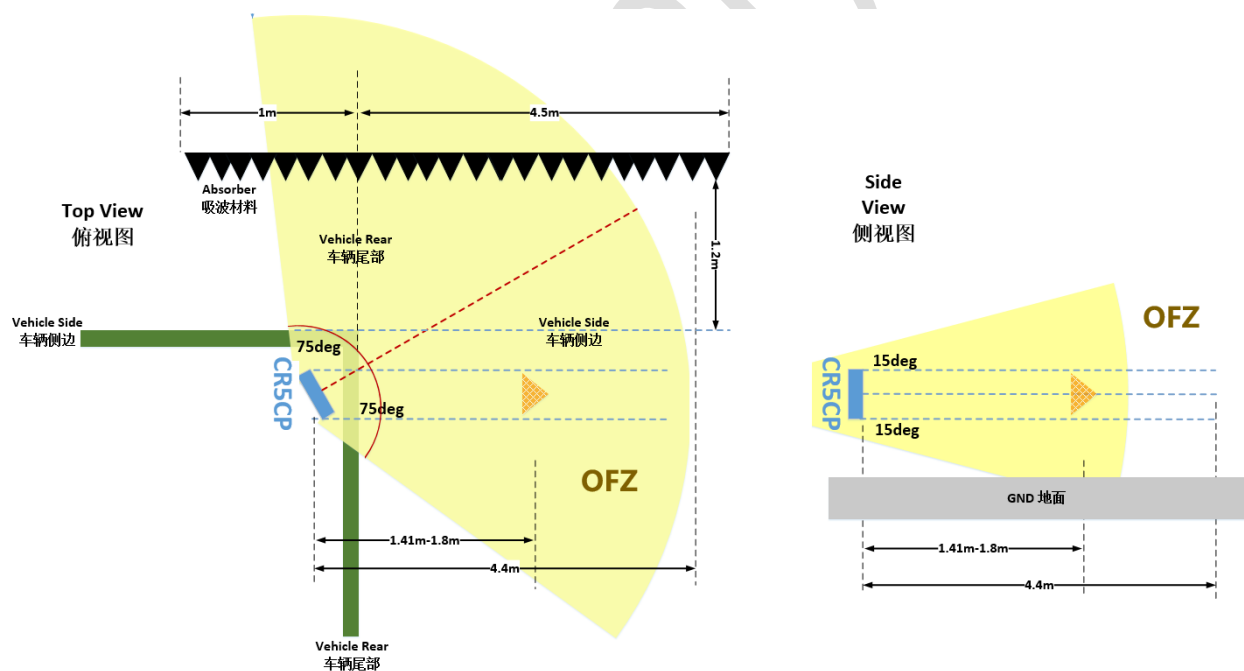
用于摆放设备及停放车辆的校准场地须平坦

The alignment area both for equipments and vehicle should be flat

在地面上不应有（或仅有有限的）金属反射物体；如果有，需要采取特殊措施来吸收来自地面的反射

No (limited) metal on the ground, if have, there must be special measure the absorber the reflection from ground.

设备摆放区域最大不平整度 maximum unevenness of area for placing alignment equipment		1 mm
车辆停放区域最大不平整度 maximum unevenness of area for vehicle		10 mm
反射避免区 Object free zone	径向 Radial	在雷达与角反射器之间 (1.4 – 1.8m) 以及角反射器后方 4.4m No obstructions between the sensor and the reflector (1.4 – 1.8m) as well as within 4.4m behind the rear bumper. Especially no obstructions or metal in this area, such as metal ground or metal shelf etc.
	垂直方向 Vertical	±15°
	水平方向 Horizontal	±75°
	其他 Others	如下注意事项 Attention as below



对吸波材料的要求：

- 吸波材料的高度 = 雷达到角反射器的距离 × $\tan 20^\circ$ + 雷达高度
- 车辆侧面吸波材料宽度为 1000mm，车辆后方吸波材料宽度为 4500mm；

- 吸波材料到车辆距离: $\geq 1.2\text{m}$;
- 吸波材料型号/品牌要求

吸波材料的型号应适用于 **76-77GHz** 频率的雷达。品牌不作要求。如有需要, 可以根据客户的要求对吸波材料样品进行测试。

Requirements for surrounding absorber:

- Height of absorber = $\text{DIST_C2T} * \tan 20^\circ + \text{height_of_radar}$;
- Width of absorbers
- side of the vehicle: **1000mm**;
- rear side of the vehicle: **4500mm**;
- Distance from absorber to vehicle: $\geq 1.2\text{m}$;
- Absorber type/Brand requirement

The absorber type shall be valid for **76-77GHz** frequency radar. The Brand is not required. The absorber sample can be tested per customer requirement.

5.3.4 车辆 Vehicle

雷达校准对车辆有如下要求:

- 确保雷达校准前车辆已做好四轮定位。
- 正确的胎压。
- 车辆稳定的停在转鼓上。
- 校准时, 车辆空载。
- 校准完成后车身重量不允许有过大变动, 从而造成车辆行驶轴线的偏转;
(例如, 在校准后进行车辆进行改造, 安装液化器罐等, 会造成行驶轴线垂直方向的角度变化)
- 驻车制动激活(可选)。

Radar alignment have requirement to vehicle as below:

- Before radar alignment, vehicle should have finished four wheel alignment.
- Correct tire pressure
- Vehicle should stay on drum stable.
- When alignment, vehicle is unload.
- no heavy parts that might change the driving axis shall be assembled after the alignment
(For example, liquid gas tank)

- Parking brake active (optional)

5.4 客户项目产线校准实际参数 Customer project EOL alignment data (依据 YYYY.MM.DD 最新参数进行计算)

角反射器 L 边长度 L value of corner reflector		XXX mm (见图 3) XXX mm (see Figure 3)
角反射器在Y方向上与后轴两侧车轮接地点中心的距离 y-distance from corner reflector to rear axle center 见图 2 See Figure 2	左前 Front Left	xx mm (同雷达安装位置) (similar to radar mounting position in y)
	右前 Front Right	
	左后 Rear Left	
	右后 Rear Right	
参考目标据雷达天线罩表面距离 Distance from radar to reflector		XXXmm(依据 customer 提供的最新数据) XXXmm(based on customer latest data)
校准参考目标中心垂直位置 Reflector center position. Vertical		雷达中心高度 (见图 4) Radar center height (see Figure 4)
雷达中心高度 Radar center height	左前 Front Left	前轴轮眉高度均值+雷达中心距轮眉高度 Average of from axes wheel housing edge height + Radar center to wheel housing edge height
	右前 Front Right	
	左后 Rear Left	
	右后 Rear Right	
前/后轮轮眉高度平均值 average wheel housing edge height of front/rear wheels		每台车通过供应商设备实测 Test by supplier equipment for each car
雷达中心距轮眉高度 The height from Radar center to wheel housing edge		需 customer 提供(need to be offered by customer)

注意:

- 如果车辆完成校准后从前方驶出, 主机厂要考虑角反射器的移动;
- 以上所有参数均基于 XX 客户提供的基本信息(YY.MM.DD)计算得出, 如实际参数与该信息不符, 风险不由博世承担相应参数须重新进行确认。

Attention:

- If vehicle move forward after EOL, OME need consider the removing of reflector.

- All parameter of the reflector is based on **XX customer** basic data and key information (**YY.MM.DD**), if the real parameter changed or different, all data need to be updated, any risk is not BOSCH responsibility.



图 4：CR5CP 中心点

Figure 4: Center of CR5CP

5.5 测量的误差 Inaccuracy of the measurement

测量的误差包括如下因素：

The error measurement include:

误差组成 Error component	水平方向 Horizontal	垂直方向 Vertical	责任方 Responsible
CR5CP installation tolerance 雷达安装误差	$\pm 3^\circ$	$\pm 3^\circ$	OEM
主动式校准的不准确度（不含温度及寿命影响） inaccuracy of active alignment measurement (w/o influence due to temperature, lifetime)	$\pm 0.5^\circ$	$\pm 0.5^\circ$	博世 Bosch
角反射器在X方向上与车辆行驶轴线的夹角的最大偏差角 Maximal deviation of angle between corner reflector normal and driving axis	$\pm 0.15^\circ$	$\pm 0.15^\circ$	OEM

注意 Note:

- 当雷达安装于保险杠、覆盖件、隔栅后方及有其他目标反射干扰时，可能还会产生额外的误差。

Additional inaccuracies may arise due to installation behind bumper, cover, grill, reflection from other objects.

- 雷达安装误差 ($\pm 3^\circ$) 是指将雷达安装在支架上，在校准之前，雷达天线轴线与设计值的夹角。

Radar installation tolerance is the deviation tolerance which between radar and bracket installation reference surface, before alignment start.

3. 请注意角反射器的法线与车辆行驶轴线（由非转向轮的前束角决定，和转向轮无关，见图 5）的偏差，博世要求行驶轴线与角反射器法线的偏差值可测，且在雷达校准之前该值必须保证在 $\pm 0.15^\circ$ 范围内。需要注意的是，如果车辆在调整自身参数（如调整前束）时改变了行驶轴线，则需要在校准开始之前重新进行转鼓，保证行驶轴线与反射板法线的偏差在 $\pm 0.15^\circ$ 范围内。

Please make sure the maximum deviation of corner reflector normal and driving axis (driving axis determined by the toe angle of non-steering wheel, see fig.5) could be measured by the alignment station, and the threshold should be within $\pm 0.15^\circ$. If driving axis has been changed during toe angle alignment or other parameter changing, the car should be revolved again on drum tester to make sure the maximum deviation of corner reflector normal and driving axis within $\pm 0.15^\circ$.

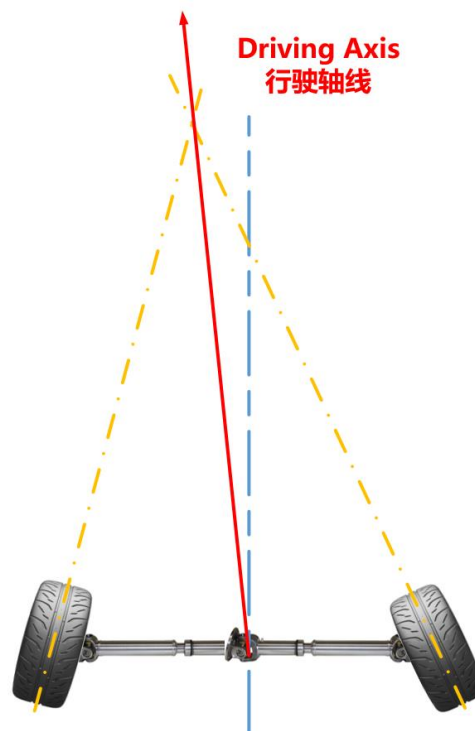


图 5：行驶轴线的定义

Figure 5: Definition of driving axis

5.6 校准的失效原因 Known limitations

如果有以下情况发生，主动式校准可能会失败

The active alignment measurement may fail or may work incorrectly if

- 角反射器没有正确放置 alignment reflector is not positioned correctly
- 反射器到雷达前方过近 alignment corner reflector is too closed to the sensor

- 向角反射器发射的雷达波被遮挡 the radar beam to the alignment reflector is blocked
- 雷达的安装偏差过大 If the sensor is excessively strongly misaligned
- 有来自其他与反射器径向距离类似的目标的反射干扰
Reflections from other objects with similar radial distance as reflector
- 在目标区域内有来自其他目标的更强烈的反射
Stronger reflection from another target inside search area
- 雷达天线特性由于反射板和雷达间其他物体的干扰发生了改变
The antenna diagram is changed by something between sensor and reflector plate

6 诊断功能 Diagnostic Functions

对于下线校准，雷达在交付客户时将附带有以下的默认配置
The CR5CP is delivered to the customer with the following coding:

- 产线模式 production mode on
- 雷达未校准 alignment never done

产线模式将雷达系统功能关闭以确保产线上不会有误触发，该模式同时将雷达波调制设置为适合校准的方式，雷达被设置为未校准状态以防止交付未校准的雷达。

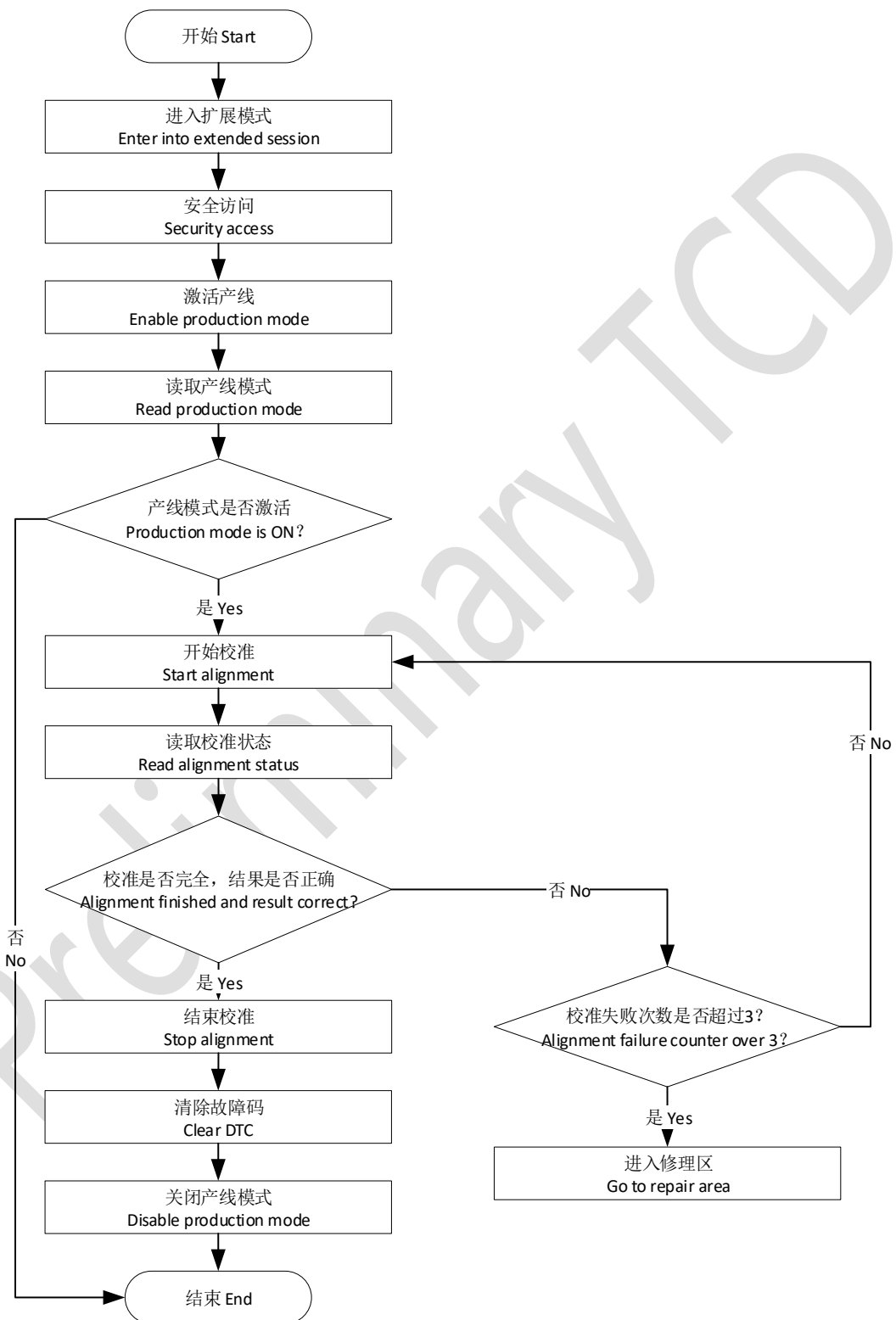
The production mode disables all functions to make sure that no false activations occur on the assembly line. This mode also sets the radar modulation suitable for alignment. The ECU is coded as misaligned as a precaution against delivering an ECU without EOL alignment.

CR5CP 的诊断码

Diagnostic ID of CR5CP

CR5CP	发送 Tx	接收 Rx
左前 Front left	TFL	RFL
右前 Front right	TFR	RFR
左后 Rear left	TRL	RRL
右后 Rear right	TRR	RRR

6.1 校准流程 Detailed process and describe



序号 No.	人工操作 Manual Operation	数 据 流 (十六进制) Data flow (HEX)	说 明 Description
1	点击开始 SDA 按钮 Click Start SDA button	XXX 02 10 03 CC CC CC CC CC XXX 06 50 03 00 32 00 C8 CC	进入扩展模式 Enter extended mode
2	/	XXX 02 27 01 CC CC CC CC CC XXX 06 67 01 Seed#0 #1 #2 #3 CC XXX 06 27 02 Key#0 #1 #2 #3 CC XXX 02 67 02 CC CC CC CC CC	安全访问 算法请参考诊断规范的 KEY-SEED 算法 Security access Reference to KEY-SEED 算法
3	/	需要定义客户所需信息 TBD by customer XX	读 ECU 的 ID Read the ID of ECU
4	/	XXX 04 2E XX XX 01 CC CC CC XXX 03 6E XX XX CC CC CC CC	打开ECU的Production Mode, 设为01 Set production mode of ECU 01
5	/	XXX 03 22 XX XX CC CC CC CC XXX 04 62 XX XX xx CC CC CC	读ECU的Production Mode, 如果xx为01则继续下面步 骤, 否则退出测试 Read the production mode of ECU, if xx is 1, then continue the following steps, or abort the test
6	探测参考目 标 Measure the refletor	XXX 04 31 01 XX XX CC CC CC XXX 04 71 01 XX XX CC CC CC	开始校准 Start alignment
7	读取校准状 态 Read alignment status	XXX 04 31 03 XX XX CC CC CC XXX 00 09 71 03 XX XX X1 X2 X3 X4 X5 CC	X1-X5 各字节含义如下: The meaning for each byte as follows: X1: Bit0-Bit3, Routine Status;

			0x00: Routine inactive; 0x01: Routine active; 0x02: Routine NVM write not OK; 0x03: Routine timeout; 0x04: Routine finished correctly; 0x05: Routine aborted; Bit4-Bit7: Routine Result; 0x00: Alignment no result available; 0x01: Alignment incorrect result; 0x02: Alignment correct result; X2 X3 表示 Horizontal deviation angle; X4 X5 表示 Vertical deviation angle. Horizontal deviation angle = $X2X3 * 0.01$; Vertical deviation angle = $X4X5 * 0.01$.
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8	停止 Stop	XXX 04 31 02 XX XX CC CC CC XXX 04 71 02 XX XX CC CC CC	停止校准 Stop alignment
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Add Finish and Repeat button to confirm adjust operation.

在操作界面上增加“完成校准”，“重复校准”按键。用于确认现场操作。

9	Finish 按钮		a.如果步骤7或8的返回值不为 00 的情况下，如果按finish按钮,则在界面进行提示:校准未完成，请重新校准，并等待repeat按钮响应(再按finish按钮无响应);
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			if step 7 or 8 return value xx or yy is not zero, when press finish button, warning : alignment not done, please repeat alignment, then waiting repeat button response(no response with finish button); b.如果步骤7和8返回值均为00, 跳转至步骤11; if step 07 and 08 both return value xx and yy is zero, when press finish button, jump to step 12
10	Repeat按钮		按repeat按钮后, 跳转至步骤6 when press repeat button, jump to step 6
11	/	XXX 04 2E 11 C0 00 CC CC CC XXX 03 6E 11 C0 CC CC CC CC	关闭ECU的Production Mode, 设为00 Set production mode of ECU 00
12	/	XXX 04 14 FF FF FF CC CC CC XXX 01 54 CC CC CC CC CC CC	清除DTC Clear DTC

注意 Warning :

- 如果雷达校准后还有转鼓测试, “**Production Mode 设置为 0**”, 需要在转鼓测试后进行, 以避免雷达功能误触发。

If there are some other tests on the drum after CR5CP alignment, the “**Production Mode set to 0**” should be done only after drum test, in case the misact of radar function.

- 打开和关闭工厂模式以及写入 VIN 码的命令可依据主机厂要求进行修改; 打开关闭工厂模式以及写入 VIN 的命令需要在扩展模式以及安全访问后进行:

Enable/disable production mode write VIN code step could be adjusted by OEM, but should be done only after enter extended mode and security access:

	->Ignition Off ->Connect diagnostic Interface ->Ignition On	XXX 02 10 03 CC CC CC CC CC XXX 06 50 03 00 32 00 C8 CC	进入扩展模式 Enter extended mode
	/	XXX 02 27 01 CC CC CC CC CC XXX 06 67 01 Seed#0 #1 #2 #3 CC	安全访问

		XXX 06 27 02 Key#0 #1 #2 #3 CC XXX 02 67 02 CC CC CC CC CC	算法请参考[诊断规范的KEY-SEED算法] Security access Reference to KEY-SEED算法
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在进入扩展模式以及安全访问后可进行工厂模式的更改以及 VIN 的写入：

After enter extended mode and security access, production mode could be changed and VIN is possible to write in the sensor:

打开工厂模式：enable production mode(function disabled)

	/	XXX 04 2E XX XX 01 CC CC CC XXX 03 6E XX XX CC CC CC CC	打开ECU的Production Mode, 设为 01 Set production mode of ECU 01
--	---	--	--

关闭工厂模式：disable production mode(function enabled)

	/	XXX 04 2E XX XX 00 CC CC CC XXX 03 6E XX XX CC CC CC CC	关闭ECU的Production Mode, (production mode设为 00) Set production mode of ECU 00
--	---	--	--

写入 VIN 码：write VIN code

	/	XXX 00 14 2E XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX CC CC	写入VIN码 Write VIN Code
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6.2 诊断测试 Diagnostic Test

第一步的目的是确保雷达在安装至车辆后，在校准前工作正常。

The reason for this first step is to ensure that the CR5CP is working correctly after vehicle assembly before moving on to alignment.

6.2.1 进入扩展诊断会话 Start Extended Diagnostic Session

进入扩展诊断会话 Start Extended Diagnostic Session	
操作员 Operator	诊断仪/雷达控制器 Tester /Radar ECU
⇒ 熄火 Ignition Off	诊断仪 Tester:
⇒ 连接诊断接口 Connect diagnostic Interface	进入扩展诊断会话 Start extended diagnostic session
⇒ 点火 Ignition On	

⇒ 等待直至雷达系统测试完成
Wait until CR5CP system test is finished

诊断仪和雷达之间的通信是基于客户定义的诊断规范完成。

The communication set up between the Tester and ECU is based on the customer diagnostics according to the corresponding specification.

通信失败 Communication not successful ⇒ 设置失败信息，中止测试 set failure message, abort test

通信成功 Communication successful ⇒ 继续 continue

6.2.2 读 ECU 的 ID Read ECU-ID

读 ECU 的 ID Read ECU-ID	
操作员 Operator	诊断仪/雷达控制器 Tester /Radar ECU
⇒ 读 ECU 标识符 Read ECU Identification	诊断仪 Tester: 请求 ECU 识别码 Request ECU Identification 雷达控制器 ECU: 输出 ECU 识别码 Output ECU Identification + customer required information

ECU标识符的目的是为了检查雷达是否与车辆相匹配，以及避免操作员将不合适的雷达装配上去。ECU标识符须经过客户及其装配线确认，该标识符须与下线检测设备软件中存储的ECU数据相比较。

为保证产品的可追溯性，需要将车辆VIN码和CR5CP的序列号对应起来。

The intention of ECU identification is checking whether CR5CP can match the vehicle or not, and preventing operator from assembling unsuitable CR5CP. The ECU identification data to be evaluated must be confirmed by the customer and its assembly plants. The ECU identification data must be compared with the in the end of line test equipment software stored desired ECU data.

To make sure every ECU can be carried up. The CR5CP series Number should be connected with VIN.

不可知的系统 Unknown System ⇒ 中止测试 Abort Test
 检测到系统 System detected ⇒ 继续 Continue

Customer project CR5CP 项目：
 读取序列号，并和 VIN 绑定，在校准之前的“最终线”进行。
Customer project C
 R5CP Project:

Read CR5CP series Number and combine VIN at the position “Final Line” which happen before CR5CP calibration.

6.2.3 产线模式 *Production Mode*

读取产线模式 Read Production Mode	
操作员 Operator	诊断仪/雷达控制器 Tester /Radar ECU
⇒ 确认 ECU 是处于产线模式 Confirm that ECU is in production mode	诊断仪 Tester: 请求 ECU 的模式 Request ECU mode 雷达控制器 ECU: 输出 ECU 的模式 Output ECU mode

其它产线模式 Other production mode ⇒ 中止测试 Abort Test

ECU 处于产线模式 ECU in production mode ⇒ 继续 Continue

6.2.4 清除/读取诊断故障码 *Clear/Read DTCs*

清除/读取诊断故障码 Clear/Read DTCs	
操作员 Operator	诊断仪/雷达控制器 Tester /Radar ECU
⇒ 清除及读取诊断故障码 Clear and Read DTCs	诊断仪 Tester: 清除诊断故障码 Clear DTCs 请求诊断故障码 Request DTCs 雷达控制器 ECU: 删除诊断故障码（包括历史故障码） Delete DTCs (including history) 输出诊断故障码 Output DTCs

在此步骤仅允许显示以下诊断故障码

At this stage, the following DTCs only may be reported:

- 产线模式激活 Production mode active
- 雷达未校准 Sensor Misaligned
- 其它需客户确定，如安装安全气囊等 To be defined by customer depending when e.g. Airbag is attached

有产线模式及雷达未校准之外的其它诊断故障码

DTCs other than Production active and Sensor misaligned ⇒ 中止测试 Abort Test

没有其它剩余的诊断故障码 No other residual DTCs ⇒ 继续 Continue